

Topic 10: Population Models, The Logistic Equation & Phase-Line Stability

Professor Leonard Master Series · Lessons 31 to 36

Course: Applied Ordinary Differential Equations (ENGR 213 Companion)

1. Whiteboard Intuition: From Malthus to Verhulst

Professor Leonard introduces population dynamics with a reality check:

"Malthusian growth says $\frac{dP}{dt} = kP \implies P(t) = P_0 e^{kt}$. That means bacteria would engulf the Earth in three days! In the real world, food runs out, space runs out, and competition kicks in. The Logistic Equation introduces a braking term $(1 - P/K)$, where K is the Carrying Capacity. When $P \ll K$, growth is exponential. When $P \rightarrow K$, growth grinds to a halt!"

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right)$$

2. Professor Leonard's Whiteboard Problem Walkthroughs

Problem 10.1: Derivation of the Logistic Solution & Inflection Point

Statement: Solve the logistic IVP $\frac{dP}{dt} = rP \left(1 - \frac{P}{K} \right)$, $P(0) = P_0$. Show that the maximum growth rate occurs at $P = K/2$.

Step-by-Step Whiteboard Solution:

- **Step 1: Separate variables:**

$$\frac{dP}{P(1 - P/K)} = r dt \implies \frac{K dP}{P(K - P)} = r dt$$

- **Step 2: Partial Fractions:**

$$\frac{K}{P(K - P)} = \frac{1}{P} + \frac{1}{K - P}$$

- **Step 3: Integrate:**

$$\int \left(\frac{1}{P} + \frac{1}{K-P} \right) dP = \int r dt$$

$$\ln |P| - \ln |K-P| = rt + C_1 \implies \ln \left| \frac{P}{K-P} \right| = rt + C_1$$

- **Step 4: Exponentiate:**

$$\frac{P}{K-P} = Ae^{rt} \quad \text{where } A = \frac{P_0}{K-P_0}$$

- **Step 5: Solve for $P(t)$:**

$$P(t) = Ae^{rt}(K-P) \implies P(1 + Ae^{rt}) = AKe^{rt}$$

$$P(t) = \frac{AKe^{rt}}{1 + Ae^{rt}} = \frac{K}{1 + A^{-1}e^{-rt}} = \frac{KP_0}{P_0 + (K-P_0)e^{-rt}}$$

- **Step 6: Maximum Growth Rate (Inflection Point):** Growth rate is $f(P) = rP - \frac{r}{K}P^2$. Maximize by differentiating with respect to P :

$$f'(P) = r - \frac{2r}{K}P = 0 \implies P = \frac{K}{2}$$

The population grows fastest when it reaches **exactly half of carrying capacity!**

Problem 10.2: Constant Harvesting & Saddle-Node Bifurcation Catastrophe

Statement: A fishery grows logistically with $r = 1$, $K = 10$, but fish are harvested at a constant rate h :

$$\frac{dP}{dt} = P \left(1 - \frac{P}{10} \right) - h$$

1. Find the critical harvesting rate h_c above which the population collapses to extinction.
2. Analyze the stability of equilibria for $h < h_c$, $h = h_c$, $h > h_c$.

Step-by-Step Whiteboard Solution:

- **Step 1: Find equilibrium points:**

$$-\frac{1}{10}P^2 + P - h = 0 \implies P^2 - 10P + 10h = 0$$

Using quadratic formula:

$$P = \frac{10 \pm \sqrt{100 - 40h}}{2} = 5 \pm \sqrt{25 - 10h}$$

- **Step 2: Bifurcation Value h_c :** The discriminant must be non-negative for real equilibria:

$$25 - 10h = 0 \implies h_c = 2.5$$

- **Step 3: Stability Analysis across Cases:**

- **Case 1: $h < 2.5$ (Sustainable):** Two equilibria $P_1 < 5 < P_2$.
 - $P_2 = 5 + \sqrt{25 - 10h}$ is **Stable Sink** (carrying capacity reduced by harvesting).
 - $P_1 = 5 - \sqrt{25 - 10h}$ is **Unstable Threshold** (if population drops below P_1 , it collapses to 0).
- **Case 2: $h = 2.5$ (Critical):** One semi-stable equilibrium at $P = 5$.
- **Case 3: $h > 2.5$ (Over-harvesting Catastrophe):** No real equilibria exist. $\frac{dP}{dt} < 0$ everywhere.
Extinction is 100% guaranteed in finite time!

Problem 10.3: The Allee Effect (Extinction Threshold)

Statement: For $\frac{dP}{dt} = -kP \left(1 - \frac{P}{M}\right) \left(1 - \frac{P}{K}\right)$ with $0 < M < K$: Classify the stability of $P = 0, M, K$.

Step-by-Step Whiteboard Solution:

- **Equilibria:** $P = 0, M, K$.
- $0 < P < M$: $\frac{dP}{dt} < 0 \implies$ population declines to 0! M is the minimum viable population!
- $M < P < K$: $\frac{dP}{dt} > 0 \implies$ population rises to K .
- $P > K$: $\frac{dP}{dt} < 0 \implies$ population drops to K .
- **Conclusion:**
 - $P = 0$: **Stable Sink**.
 - $P = M$: **Unstable Source** (The Allee Threshold).
 - $P = K$: **Stable Sink** (Carrying Capacity).

3. Common Exam Traps & Professor Leonard Warnings

- **Trap 1: Partial Fractions Minus Sign:** $\int \frac{1}{K-P} dP = -\ln|K - P|$. Forgetting the minus sign ruins the exponent!
- **Trap 2: Confusing Rate with Population:** $P = K/2$ is the point of maximum
growth rate

, not maximum population.